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W1M1 2c confidence

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So, I have some questions.

According to the delta theorem, in the limit...

If $\hat{\theta} \sim \mathcal{N}(\theta, \sigma^2)$ then $f(\hat{\theta}) \sim \mathcal{N}(f(\theta), [f'(\theta)]^2 \sigma^2)$

In course, we saw that $\hat{\theta}_{ML} \sim \mathcal{N}(\theta, I^{-1}(\theta))$.

Hence, if we couple them, we have a confidence interval given accordingly.

$$f(\hat{\theta}) \pm z_{1-\alpha/2} f'(\hat{\theta}) \hat{\sigma}$$

However, the answer to the exercise is...

$$\exp(\hat{\theta} \pm z_{1-\alpha/2} \hat{\sigma})$$

Note that sigma above is $\hat{\sigma}_i \Delta x$ where the sigma_i means the sd error for the variable.

In real life, in case I am doing a project/research paper and need this, which method should I use?

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